



# AM RADIO RECEIVER

A radio receiver is a device which reproduces the modulated or radio waves into sound waves. The receiver has to perform following actions:

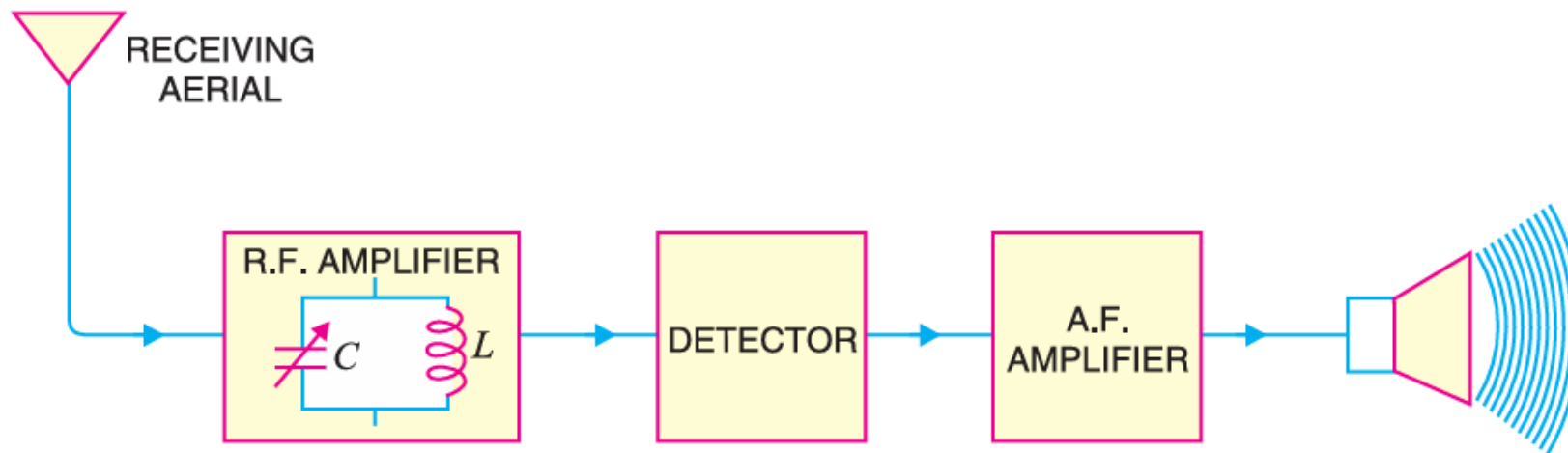
1. Receive/intercept the passing radio waves
2. Select the right signal by passing through LC circuit
3. Must amplify the received signal
4. Retrieve the audio signal from amplified radio wave
5. Amplify the audio signal and feed to speaker

AM radio receiver is broadly classified into two types:

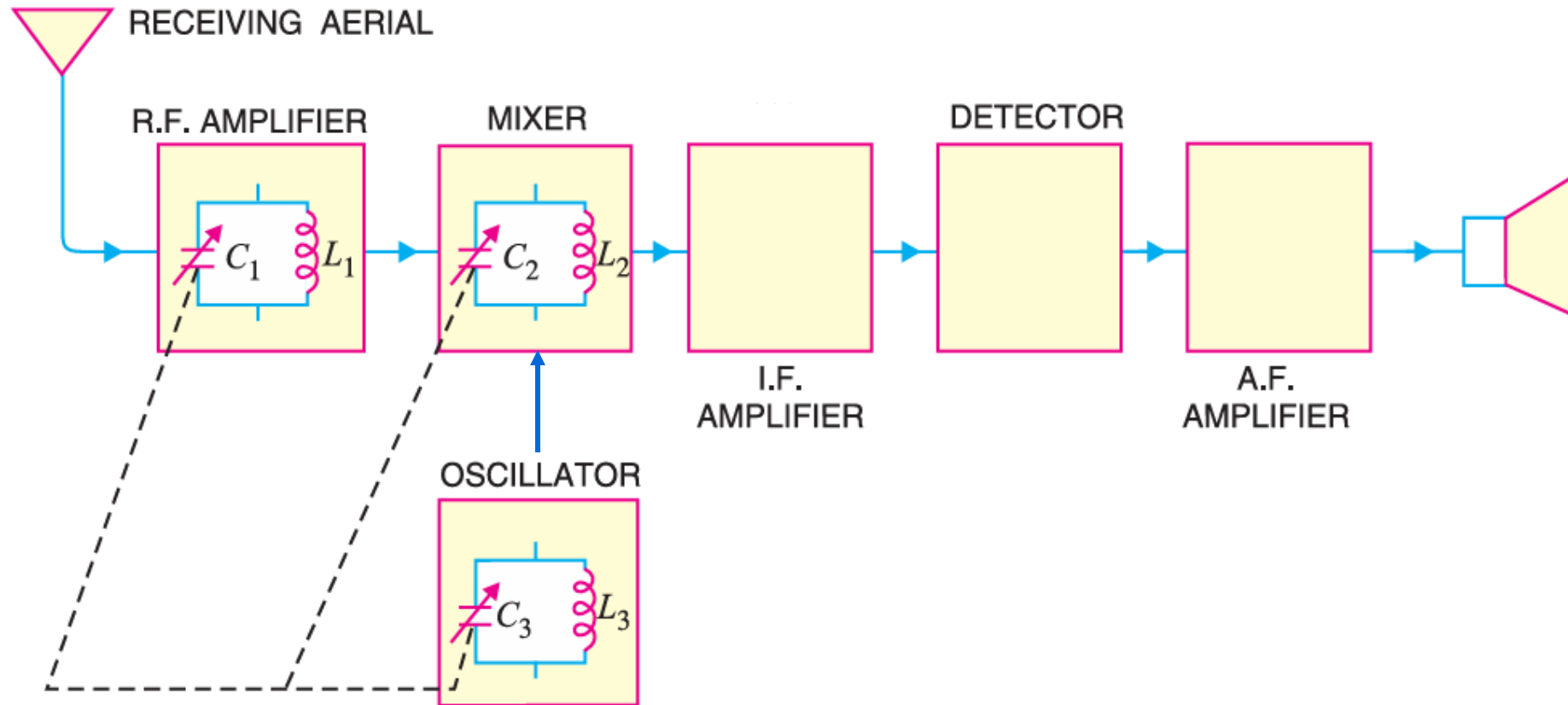
- I. Straight radio receiver
- II. Superhetrodyne receiver

# STRAIGHT RADIO RECEIVER

- Receiver intercepts radio waves from different broadcasting stations.
- The radio wave is amplified by the tuned RF amplifier.
- This amplified signal is fed to the detector circuit which extracts the audio signal from the radio wave.
- Then output goes through audio amplification.
- At last amplified audio signal is fed to the speaker for sound reproduction.
- But its limitation is that the circuit needs to be tuned for desired station.



# SUPERHETRODYNE RECEIVER



# SUPERHETRODYNE RECEIVER

Stages of superheterodyne receiver:

1. **RF amplifier circuit:** The R.F. amplifier stage uses a tuned parallel circuit L1C1 with a variable capacitor C1. The radio waves from various broadcasting stations are intercepted by the receiving aerial and are coupled to this stage. This stage selects the desired radio wave and raises the strength of the wave to the desired level.
2. **Mixer stage:** The amplified output of R.F. amplifier is fed to the mixer stage where it is combined with the output of a local oscillator. The two frequencies beat together and produce an intermediate frequency (IF), common design choice is 455 kHz. The intermediate frequency is the difference between oscillator frequency and radio frequency i.e.  $I.F. = \text{Oscillator frequency} - \text{Radio frequency}$
3. **IF amplifier stage:** Output of the previous stage is fed to amplifier that amplifies the tuned signal.
4. **Detector stage:** The output from the last IF amplifier stage is coupled to the input of the detector stage. Here, the audio signal is extracted from the IF output. Usually, diode detector circuit is used because of its low distortion and excellent audio fidelity.
5. **AF amplifier stage:** The audio signal output of detector stage is fed to a multistage audio amplifier. Here, the signal is amplified until it is sufficiently strong to drive the speaker. The speaker converts the audio signal into sound waves corresponding to the original sound at the broadcasting station

# SUPERHETRODYNE RECEIVER

## Advantages:

- ❑ The superhetrodyne principle makes it possible to produce an intermediate frequency which is much less than the radio frequency and stable, enabling high RF amplification.
- ❑ Decreases loss compared to straight receiver
- ❑ It permits use of fixed RF amplifiers hence cheaper than other radio receivers
- ❑ Better noise rejection
- ❑ Bandwidth depends on IF signal so, uniform reception and same audion quality for all stations

# REFERENCE

Principles of Electronics by VK Mehta  
Chapter- 16



**THANK YOU**

